Physical Chemistry
Thermodynamics

The Maxwell relations for closed reversible systems in which only PV-work is considered can be derived from the differential energy expressions involving: dE, dH, dG and dA.

(1) Beginning with the first law of thermodynamics, derive the equation for dE, dH, dG and dA.

(2) Using the equation that you derived for dA, obtain $(\delta A/\delta T)_V$ and $(\delta A/\delta V)_T$. Use these results to obtain the equation for $(\delta S/\delta V)_T$.

There are several Maxwell relations because of the number of ways that four varibles (T,S, P and V) can be distributed in the following manner

$$(\delta W/\delta X)_Z = \pm (\delta Y/\delta Z)_X$$

It is possible to construct any one of the relations from this equation if the following rules are used: (1) the cross products XY and WZ must be work terms (i.e., PV and ST); (2) the negative sign is chosen if W and X or Z and Y are T and V.

(3) Obtain the four Maxwell relations for:

$$(\delta T/\delta V)_S =$$

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$$(\delta S/\delta V)_T =$$

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